

HW10_Jones

Install the Packages.

```
library(ggplot2)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v lubridate  1.9.4      v tibble     3.3.0
v purrr      1.1.0      v tidyr      1.3.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
```

```
# install.packages("kableExtra") # run once
library(knitr)
library(kableExtra)
```

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

group_rows

Create a frequency table for democ_regime

```
world.data <- read.csv("world.csv")
table(world.data$democ_regime)
```

```
No Yes
75 114
```

Create a frequency table for typerel variable

```
table(world.data$typerel)
```

eastern	Hindu	Jewish	Muslim
15	2	1	50
Orthodox	other	Protestant	Roman Catholic
13	12	35	63

Make the typerel frequency table vertical using data.frame

```
data.frame(table(world.data$typerel))
```

	Var1	Freq
1	eastern	15
2	Hindu	2
3	Jewish	1
4	Muslim	50
5	Orthodox	13
6	other	12
7	Protestant	35
8	Roman Catholic	63

Create a complete frequency table for typerel with percentages

```
ft.typerel <- data.frame(table(world.data$typerel))
ft.typerel$Percent <- round(prop.table(ft.typerel$Freq) * 100, digits = 2)
colnames(ft.typerel)[colnames(ft.typerel) == "Var1"] <- "Religion"
```

Display the table

Religion	Freq	Percent
eastern	15	7.85
Hindu	2	1.05
Jewish	1	0.52
Muslim	50	26.18
Orthodox	13	6.81
other	12	6.28
Protestant	35	18.32
Roman Catholic	63	32.98

```
ft.typerel
```

```

      Religion Freq Percent
1      eastern   15    7.85
2        Hindu    2    1.05
3       Jewish    1    0.52
4       Muslim   50   26.18
5    Orthodox   13    6.81
6        other   12    6.28
7   Protestant   35   18.32
8 Roman Catholic  63   32.98

```

```
kbl(ft.typerel, booktabs = TRUE) |>
  kable_styling(latex_options = "scale_down")
```

Which religion is the most “popular” in the world? ##Christian, as it is the mode of the typerel variable and has the highest frequency.

What is the percentage of countries where muslim is the majority? ANswer: 26.18%

Create a frequency table for Democ_regime with percentages.

```
ft.democ <- data.frame(table(world.data$democ_regime))
ft.democ$Percent <- round(prop.table(ft.democ$Freq) * 100, digits = 2)
colnames(ft.democ)[colnames(ft.democ) == "Var1"] <- "Regime_Type"
ft.democ
```

```

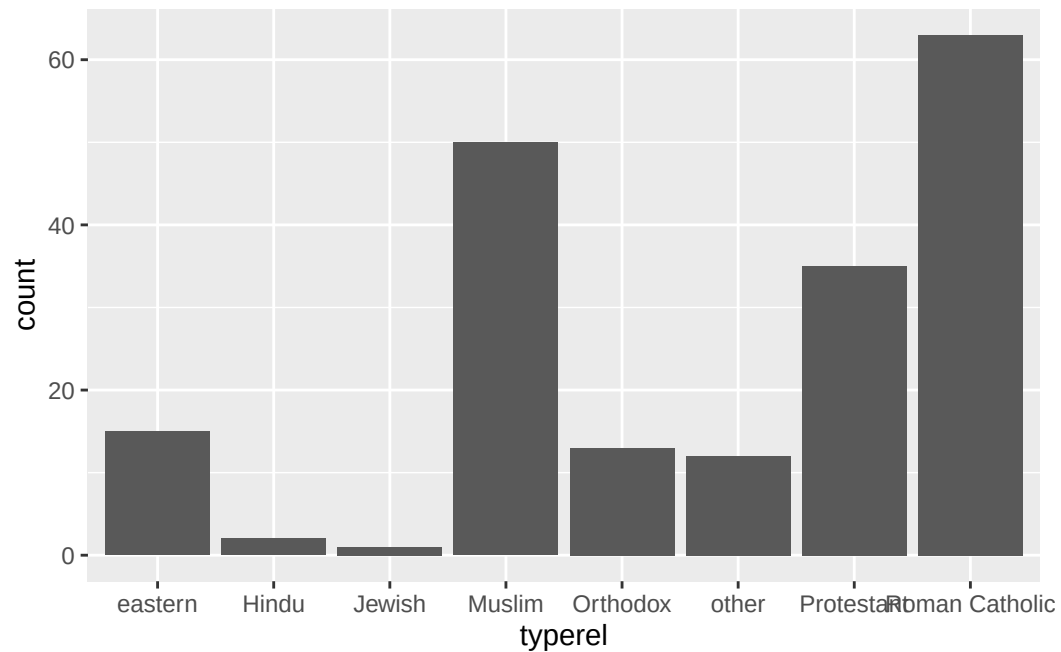
Regime_Type Freq Percent
1          No   75   39.68
2          Yes  114   60.32

```

What percentage of countries have a democratic regime? Answer: 60.32

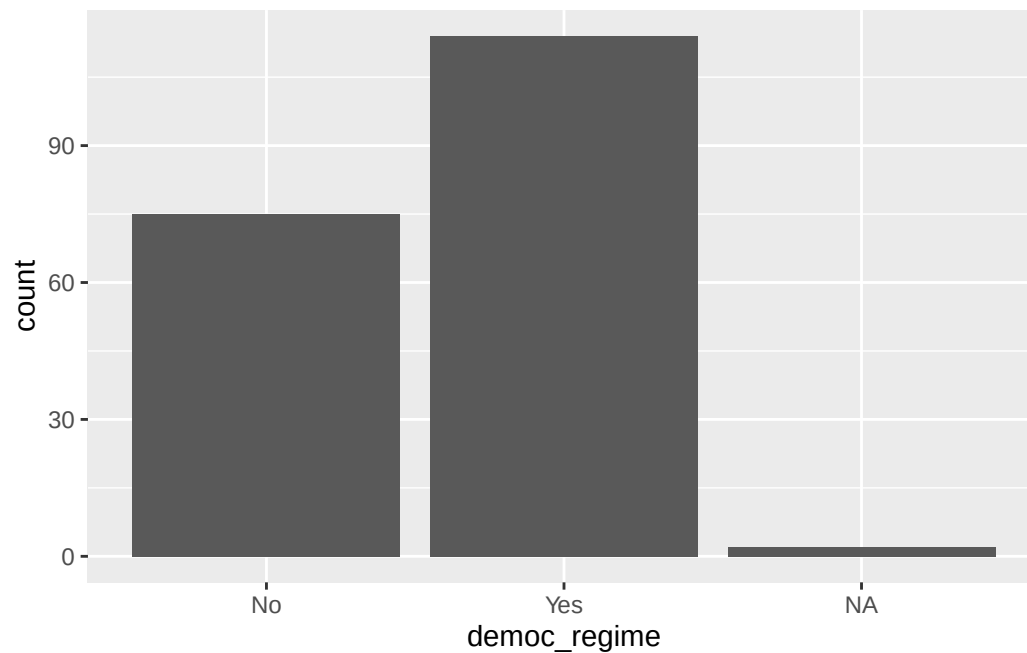
Create a bar chart to summarize the typerel variable

```
ggplot(world.data, aes(x = typerel)) + geom_bar()
```



Create a bar chart to summarize the democ_regime variable

```
ggplot(world.data, aes(x = democ_regime)) + geom_bar()
```



Create a bar chart for typerel variable and save as a PDF

```
g <- ggplot(world.data)
g <- g + aes(x = typerel)
g <- g + geom_bar()
g <- xlab("Predominant Religion")
g <- ylab("Number of Countries")
g <- g + theme(axis.text.x = element_text(size = 12, angle = 45, hjust = 1))
g
```

NULL

```
ggsave(file = "typerel_bar.pdf", width = 10, height = 8)
```

Numerically summarize the gini04 variable.

Measures of central tendency for gini04 variable (mean and median)

```
mean(world.data$gini04, na.rm = TRUE)
```

```
[1] 40.13889
```

```
median(world.data$gini04, na.rm = TRUE)
```

```
[1] 37.95
```

Measures of dispersion for gini04

```
sd(world.data$gini04, na.rm = TRUE)
```

```
[1] 10.35998
```

```
var(world.data$gini04, na.rm = TRUE)
```

```
[1] 107.3291
```

```
range(world.data$gini04, na.rm = TRUE)
```

```
[1] 24.4 70.7
```

```
IQR(world.data$gini04, na.rm = TRUE)
```

```
[1] 14.45
```

Complete summary for gini04 variable

```
summary(world.data$gini04)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
24.40	32.42	37.95	40.14	46.88	70.70	65

Numerically summarize the gini08 variable. #Measures of central tendency for gini08 variable (mean and median)

```
mean(world.data$gini08, na.rm = TRUE)
```

```
[1] 40.74252
```

```
median(world.data$gini08, na.rm = TRUE)
```

```
[1] 39.2
```

Measures of dispersion for gini08

```
sd(world.data$gini08, na.rm = TRUE)
```

```
[1] 9.999242
```

```
var(world.data$gini08, na.rm = TRUE)
```

```
[1] 99.98484
```

```
range(world.data$gini08, na.rm = TRUE)
```

```
[1] 24.7 74.3
```

```
IQR(world.data$gini08, na.rm = TRUE)
```

```
[1] 13.55
```

Complete summary for gini08 variable

```
summary(world.data$gini08)
```

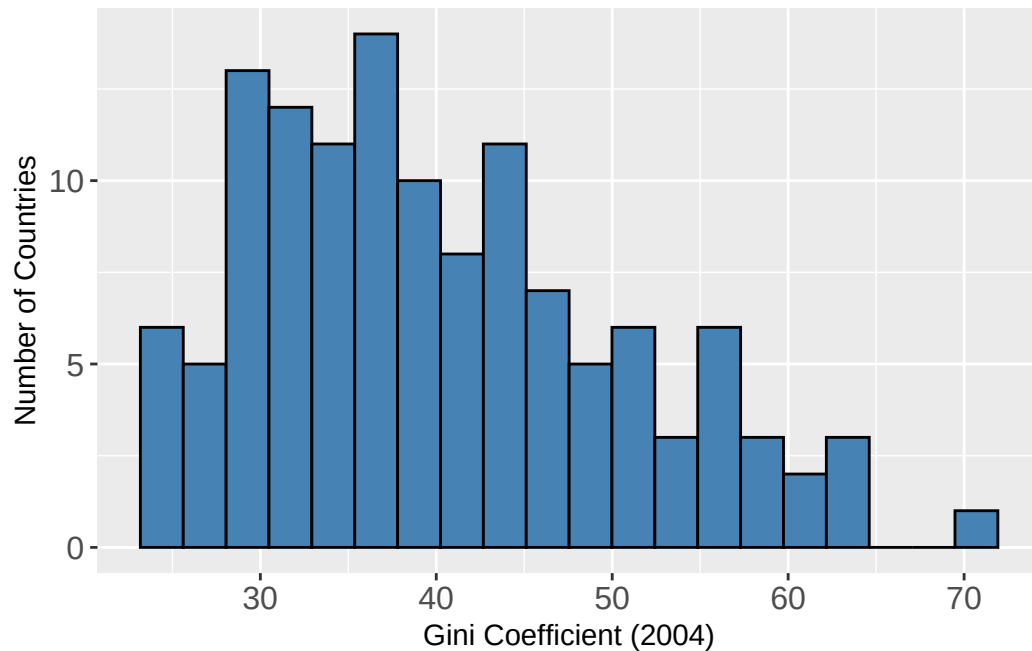
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
24.70	33.55	39.20	40.74	47.10	74.30	64

Comparing the distributions of gini04 and gini08 #To see if economic inequality is getting better or worse, I would choose to compare the mean and medians of both variables. If the mean and median are higher for gini08 over gini04, then inequality is getting worse. #If they are lower, however, inequality is getting better. I think it would also help to look at the standard deviations to see if the dispersion has changed in any way. In comparing the ranges and IQRs, this can tell us if the extremes have either increased or decreased.

Create a histogram of gini04

```
g1 <- ggplot(world.data)
g1 <- g1 + aes(x = gini04)
g1 <- g1 + geom_histogram(bins = 20, fill = "steelblue", color = "black")
g1 <- g1 + xlab("Gini Coefficient (2004)")
g1 <- g1 + ylab("Number of Countries")
g1 <- g1 + theme(axis.text = element_text(size = 12))
g1
```

Warning: Removed 65 rows containing non-finite outside the scale range (``stat_bin()``).



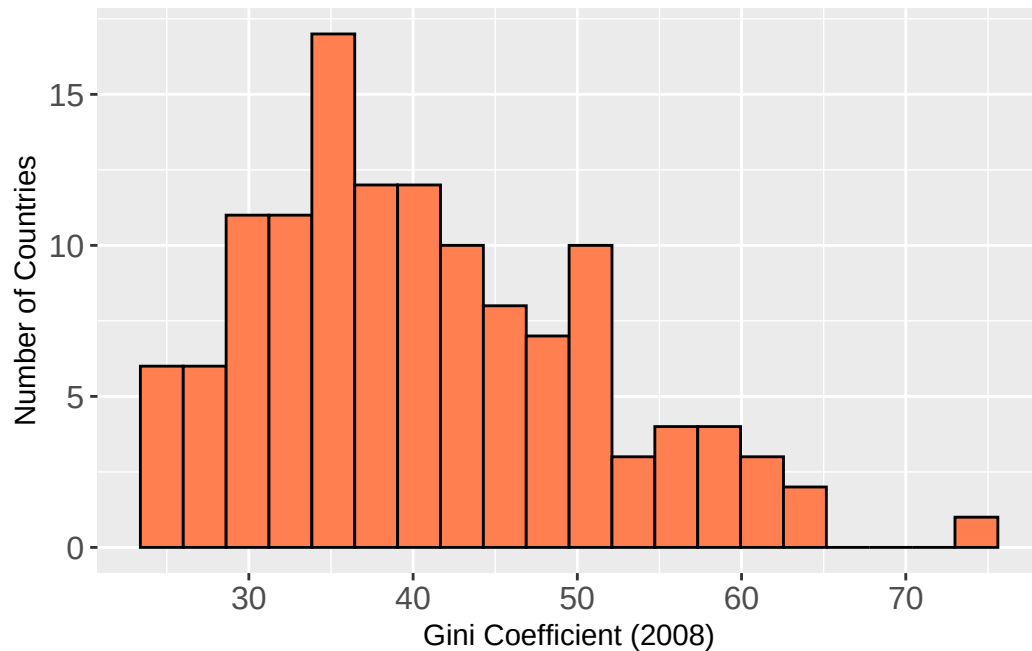
```
ggsave(file = "gini04_histogram.pdf", width = 10, height = 8)
```

Warning: Removed 65 rows containing non-finite outside the scale range (`stat\_bin()`).

Create a histogram of gini08

```
g1 <- ggplot(world.data)
g1 <- g1 + aes(x = gini08)
g1 <- g1 + geom_histogram(bins = 20, fill = "coral", color = "black")
g1 <- g1 + xlab("Gini Coefficient (2008)")
g1 <- g1 + ylab("Number of Countries")
g1 <- g1 + theme(axis.text = element_text(size = 12))
g1
```

Warning: Removed 64 rows containing non-finite outside the scale range (`stat\_bin()`).



```
ggsave(file = "gini08_histogram.pdf", width = 10, height = 8)
```

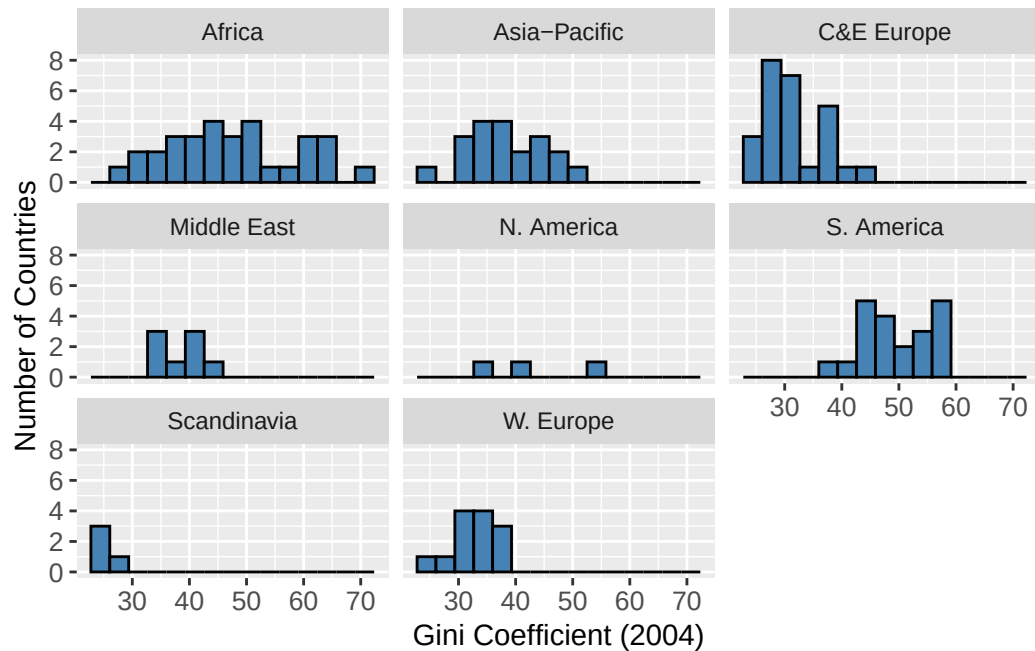
Warning: Removed 64 rows containing non-finite outside the scale range (``stat_bin()``).

Comparing the graphical distributions #With the two graphs next to each other, comparing the shapes, spreads of distribution, and centers becomes much easier. If the gini08 histogram is shifted to the right in comparison with the gini04, it again shows that inequality has increased. The shapes of distribution also show the skewness and if there are more countries with extremely inequality.

Create separate histograms of gini04 for different regions

```
g3 <- ggplot(world.data)
g3 <- g3 + aes(x = gini04)
g3 <- g3 + geom_histogram(bins = 15, fill = "steelblue", color = "black")
g3 <- g3 + xlab("Gini Coefficient (2004)")
g3 <- g3 + ylab("Number of Countries")
g3 <- g3 + facet_wrap(~ region)
g3<- g3 + theme(axis.text = element_text(size = 10))
g3
```

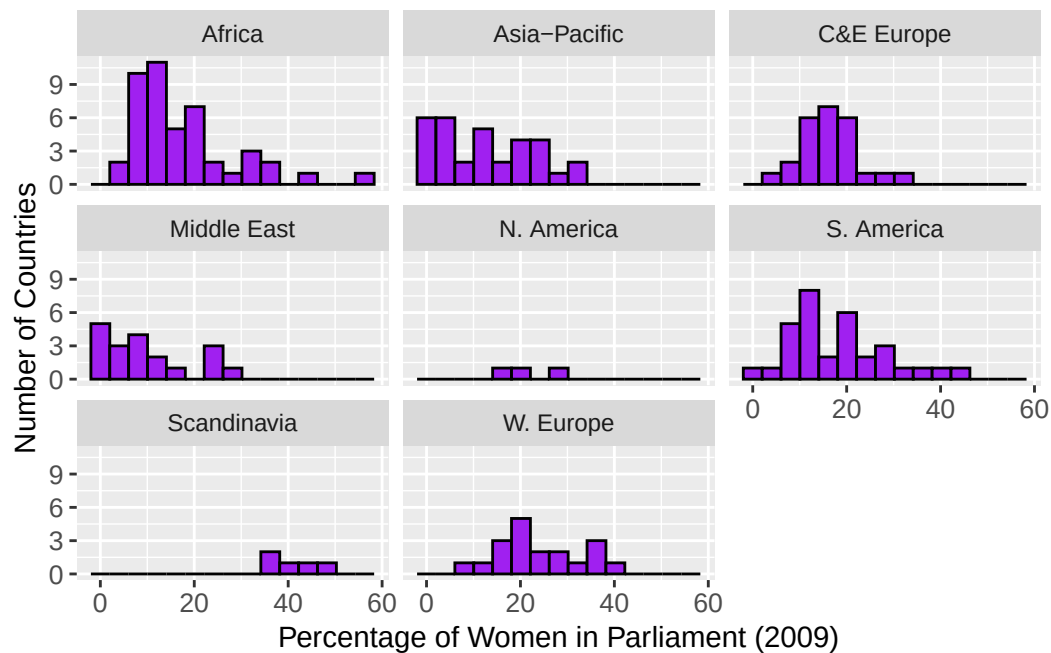
Warning: Removed 65 rows containing non-finite outside the scale range (``stat_bin()``).



Create separate histograms of women09 for different regions

```
g4 <- ggplot(world.data)
g4 <- g4 + aes(x = women09)
g4 <- g4 + geom_histogram(bins = 15, fill = "purple", color = "black")
g4 <- g4 + xlab("Percentage of Women in Parliament (2009)")
g4 <- g4 + ylab("Number of Countries")
g4 <- g4 + facet_wrap(~ region)
g4 <- g4 + theme(axis.text = element_text(size = 10))
g4
```

Warning: Removed 11 rows containing non-finite outside the scale range (`stat\_bin()`).



Obtain numerical summaries of gini04 for different regions

```
by(world.data$gini04, world.data$region, summary)
```

world.data\$region: Africa

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
28.90	39.30	47.00	47.31	54.70	70.70	18

world.data\$region: Asia-Pacific

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
24.90	33.98	36.45	38.23	43.40	50.90	17

world.data\$region: C&E Europe

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
24.40	28.25	30.85	31.77	35.83	45.60	1

world.data\$region: Middle East

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
33.40	35.23	37.95	37.75	39.85	43.00	11

world.data\$region: N. America

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
33.10	36.95	40.80	42.83	47.70	54.60

world.data\$region: S. America

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
37.90	44.60	49.10	49.55	55.10	59.10	11

```
-----
world.data$region: Scandinavia
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
 24.70  24.93   25.40   25.60  26.07   26.90     1
-----
```

```
world.data$region: W. Europe
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
 25.00  30.80   32.70   32.83  35.90   38.50     6
```

Calculate standard deviation of gini04 for different regions

```
by(world.data$gini04, world.data$region, sd, na.rm = TRUE)
```

```
world.data$region: Africa
[1] 11.06417
-----
```

```
world.data$region: Asia-Pacific
[1] 6.651417
-----
```

```
world.data$region: C&E Europe
[1] 5.167651
-----
```

```
world.data$region: Middle East
[1] 3.314901
-----
```

```
world.data$region: N. America
[1] 10.89327
-----
```

```
world.data$region: S. America
[1] 6.329504
-----
```

```
world.data$region: Scandinavia
[1] 0.9831921
-----
```

```
world.data$region: W. Europe
[1] 3.679761
```

Which region has the smallest dispersion? It looks like either Western Europe or Scandinavia due to the regions having more similar (and usually lower) levels of inequality across countries.